Cement and Concrete Composites 109 (2020) 103564

Contents lists available at ScienceDirect

Cement and Concrete Composites

journal homepage: http://www.elsevier.com/locate/cemconcomp

Comparative durability of GFRP composite reinforcing bars in concrete and in simulated concrete environments

Allan Manalo [a,*], Ginghis Maranan [a], Brahim Benmokrane [b], Patrice Cousin [b], Omar Alajarmeh [a], Wahid Ferdous [a], Ray Liang [c], Gangarao Hota [c]

[a] Centre for Future Materials (CFM), School of Civil Engineering and Surveying, University of Southern Queensland, Toowoomba, QLD, 4350, Australia
[b] University of Sherbrooke, Department of Civil Engineering, Sherbrooke, Quebec, Canada
[c] Department of Civil and Environmental Engineering, West Virginia University, Morgantown, WV, 26506, USA

ARTICLE INFO

Keywords:
GFRP composite bars
Concrete embedded bars
Interlaminar shear strength
Comparative durability
Alkaline solution
Time-shift factor

ABSTRACT

Many studies suggest that the durability of glass-fiber-reinforced-polymer (GFRP) bars in a simulated concrete pore solution is very different than the same bars in an actual concrete environment. This study conducted a comparative evaluation of the durability of GFRP bars in concrete and in simulated concrete environments by investigating their interlaminar shear strength. It focused on evaluating the physical, mechanical, and micro-structural properties of GFRP bars subjected to high moisture, saline, and alkaline environments. Bare GFRP bars and cement-embedded GFRP bars were immersed in solutions at different temperatures (23 °C, 60 °C, and 80 °C) and for different exposure times (28, 56, and 112 days). The results show that the percentage water uptake and the apparent diffusivity of the GFRP bars were strongly dependent on the type and temperature of the immersion solution. The interlaminar shear strength of the GFRP bars directly immersed in a solution degraded more significantly than those embedded in concrete and immersed. Moreover, the alkaline solution was more aggressive to the GFRP bars than tap water or saline solution, affecting bar fiber, matrix interface, and chemical structure. As a result of this study, master curves and time-shift factors were developed to correlate the retention of interlaminar shear strength from the accelerated aging test to the service life of GFRP bars in an actual concrete environment.

1. Introduction

Glass-fiber-reinforced-polymer (GFRP) bars have emerged as a promising and cost-effective replacement for steel to increase the useful life of reinforced-concrete structures exposed to severely aggressive environments. Extensive research and development has been conducted in the US, Canada, Europe, and Japan [1] on this reinforcing material, which has a number of characteristics, such as high tensile strength; light weight; noncorroding, nonmagnetic, and nonconductive properties. This work has led to the many successful field applications of GFRP-reinforced concrete structures, including highway bridges and barriers, pavements and parking garages, storage facilities for chemical and wastewater-treatment plants, magnetic-resonance-imaging facilities, detector loops in railway lines, and temporary structures, such as soft-eyes in underground excavations and tunneling works. Composite reinforcing material is now also being increasingly investigated and used in Australia [2–6] as the main reinforcement in concrete structures near the coastline and in aggressive soils. As internal reinforcement, GFRP bars are continuously subjected to alkaline environments provided by the surrounding concrete and other environmental conditions that might affect the physical, mechanical, and long-term durability properties of the bars. Many researchers have suggested that the high alkalinity of concrete pore solutions constitutes an aggressive environment for GFRP bars, which could damage the glass fibers and/or degrade the fiber–resin interface [7–11]. Similarly, infrastructure systems are exposed to external agents during their service life, including high moisture, alkalinity, and saline environments that can degrade the material properties [12] and the bond of GFRP bars in concrete [13,14]. Given the limited understanding about GFRP-bar durability, Ceroni et al. [15] and Karbhari and Zhang [9] indicated that designers apply very conservative safety factors to account for the unquantified detrimental effects which eliminates the high-strength performance of this reinforcing material. Thus, Micelli and Nanni [16] and Gooranorimi and Nanni [17] highlighted the importance of understanding the long-term

* Corresponding author.
E-mail address: allan.manalo@usq.edu.au (A. Manalo).

https://doi.org/10.1016/j.cemconcomp.2020.103564
Received 26 July 2019; Received in revised form 24 January 2020; Accepted 12 February 2020
Available online 20 February 2020

and durability performance of GFRP bars under different aggressive environmental conditions, since this performance is critical to the widespread acceptance of GFRP reinforcement for civil-infrastructure applications.

In recent years, significant efforts have gone into studying the effects of highly aggressive environments on GFRP-bar durability. Most studies have focused on high moisture, saline, and alkaline environments as these types of environments are considered environmental problems in reinforced-concrete elements [15]. In most cases, the durability of this reinforcing material is determined based on the changes in bar mechanical properties as the result of accelerated testing and evaluation programs using bare GFRP bars [18]. Kim et al. [19] exposed GFRP bars directly to different solutions at room and elevated temperatures to accelerate degradation. They found that the alkaline solution reduced the tensile strength of E-glass/vinyl-ester FRP bars by almost 60% after 132 days. Based on the model presented by Davalos et al. [20], GFRP bars made of E-glass and vinyl-ester resin can retain only 38% of their tensile strength after 50 years of exposure in saturated loaded concrete at 10 °C. In most of these durability studies, the GFRP bars experienced significant loss of mechanical properties and it was impossible to make full use their superior properties. Almusallam et al. [21], however, highlighted that the accelerated laboratory experiments were harsher than actual field conditions. Robert et al. (2009) found that GFRP bars embedded in moist concrete and exposed to tap water at elevated temperature could retain up to 90% of their tensile strength after 240 days of exposure. Moreover, in-field durability testing indicated that there was no degradation of the GFRP bars in the concrete structures. Mufti et al. [23] found no degradation of the GFRP bars in five concrete bridge structures across Canada after exposure to natural environmental conditions for 5–8 years. Gooranorimi and Nanni [17] further validated the long-term durability of GFRP bars extracted from the concrete deck of the Sierra de la Cruz Creek Bridge in Texas (US) after 15 years of service. More recently, Benmokrane et al. [24] reported no significant changes in the physico-chemical properties and microstructure of GFRP bars extracted from a concrete bridge barrier on the Val-Alain Bridge in Canada after 11 years of service exposure to wet–dry cycles, freeze–thaw cycles, and deicing salts. Mufti et al. [23] pointed out that the durability results for GFRP bars in actual concrete structures are very different from those produced by accelerated laboratory testing in which the bars are immersed in alkaline solution because the concrete itself shields the GFRP bars from direct exposure to various environmental conditions. Thus, a more realistic, simple study should be conducted to better understand the durability of GFRP bars in concrete environments. Moreover, predicting the service life and long-term performance of GFRP bars subjected to different environmental factors is of immense importance to the further use of these non-corrodible reinforcing materials.

The above studies clearly indicate that the accelerated testing of bare GFRP bars subjected to simulated concrete pore solutions yields very different durability results than those of bars in actual concrete environments over a long period of time. In fact, D'Antino et al. [25] highlighted that the environmental aging of GFRP bars by different research groups often produced contradictory results. Most of these studies were conducted by characterization of the tensile-test properties of GFRP bars, flexural investigation of concrete beams reinforced with GFRP bars subjected to accelerated aging testing, or extraction of GFRP bars from actual concrete structures for in-field service evaluation. These investigations required significant resources and time, thereby limiting the test parameters and data obtained to evaluate GFRP-bar durability. Similarly, most durability investigations focusing on tensile-strength tests measured very little change because this property is mostly defined by fiber mechanical properties [26,27]. Ceroni et al. [15] and Kim et al. [19] highlighted that accelerated aging involves exposure to moisture and elevated temperatures, which affects resin properties more than fiber properties. Field conditions limit the length of bar extraction from existing concrete structures. Consequently, ILSS characterization using the short-beam shear test method might be the only feasible option. Adams [28] indicated that the short-beam shear test is one of the most important mechanical tests for composites and an excellent choice for comparative testing. This is due to the test-method's simplicity, but it measures the interior integrity of bars, especially fiber–matrix adhesion. Thus, evaluating the interface property of GFRP bars using the short-beam shear test can give a straightforward, reliable indication of the resistance of the fiber–matrix interface after exposure to aggressive environments.

Karbhari and Zhang [9] suggested that glass fibers and vinyl-ester resin systems are preferred for civil infrastructure due to cost considerations and ease of processing. Tannous and Saadatmanesh [8] reported that vinyl-ester-based GFRP bars offer higher fiber protection and provide high resistance to chemical attack. Moreover, Benmokrane et al. [1,11] found that the vinyl-ester-based GFRP bars tested could retain almost all their original tensile strength and stiffness properties, even after long-term exposure to alkaline solutions. Based on the short-term test results for concrete beams immersed in tap water in temperature-controlled tanks, Davalos et al. [20] found that the dominant degradation mechanism for the GFRP bars was deterioration of the fiber–matrix interface. Wang et al. [29] also indicated that the fiber–resin interface is more generally easily destroyed by the aggressive solution and is commonly believed to be the weakest component in composite materials. Micelli and Nanni [16] suggested that the short-beam shear test could yield good representative results in evaluating the fiber–resin interface of GFRP bars, and these results could provide indications of potential effects on longitudinal properties. This is due to the interlaminar shear strength (ILSS) of GFRP bars, which is primarily related to resin properties and governed by the fiber–matrix interface [1]. Furthermore, Benmokrane et al. [11] suggested that the fiber–resin interface was one of the important issues in manufacturing GFRP bars. As a result, this bar property was added to the recently approved CSA S807 [30] as a new test requirement for quality-assurance testing. Therefore, variations in interlaminar shear strength could be a good indicator of deterioration in GFRP bars [15] and provide a measurement of resin damage caused by fluid penetration, which occurs during bar aging [19]. Despite its simplicity, few investigations of the durability of GFRP bars using ILSS have been conducted [1,11,15,19]; none have explored ILSS for evaluating the durability of GFRP bars under accelerated aging conditions and exposed to simulated concrete environments.

The current study provides a comparative evaluation of the durability of GFRP bars in concrete and in simulated concrete environments by investigating their interlaminar shear strength. It focuses on the evaluation of the physical, mechanical, and microstructural properties of GFRP bars in high moisture, saline, and alkaline environments. These types of environments were selected as they are considered environmental problems in reinforced-concrete elements [15]. The results from this study provide a better understanding of the durability and long-term performance of GFRP bars for safe design and application as internal reinforcement in concrete structures. It also provides a prediction of the service life of GFRP bars in an actual concrete environment based on the temperature time-shift factor determined from accelerated aging tests.

2. Experimental program

2.1. Materials

Sand-coated Grade III GFRP bars [30] with a nominal diameter (d_b) of 9.53 mm were used in the study. These GFRP bars were considered as they have been successfully applied in many infrastructure projects and are classified as highly durable reinforcement by the CSA S807 [30]. The bars were manufactured using ECR-glass fibers in a modified vinyl-ester resin in a pultrusion process. Small bar size was considered in this study due to faster moisture absorption compared to larger bar sizes [1]. The glass-fiber content by weight, determined in accordance with ISO 1172:1996(E) [31], was 84.05%. The physical and interlaminar shear properties of these bars were determined using the appropriate CSA and ASTM test standards, as reported in Table 1.

2.2. Test specimens

A total of 324 specimens were prepared, conditioned, and tested. The bars were divided into two series: (1) GFRP bars and (2) GFRP bars embedded in concrete, as shown in Fig. 1. The test specimens were cut into 40 mm lengths (approximately four times the bar diameter, $4d_b$) from GFRP bars from the same production lot. The GFRP bars embedded in concrete (Fig. 1b) were placed centrally in a PVC pipe 60 mm in length and 25 mm in diameter, which was filled with concrete to provide a cover of 7.5 mm around the bar and 10 mm at both ends. The concrete-embedded GFRP bars were prepared to simulate the environmental conditions of reinforcement in concrete structures. It should be noted that the interface between reinforcement and concrete is mostly thin cement paste with fine aggregate, which was simulated in this study. The cover thickness was kept to a minimum for easy bar removal after conditioning. The concrete used in our study consisted of a cement paste made of type 1 cement (according to ASTM C150 [39]) and a water-to-cement ratio of 0.45. Six (6) replicates were prepared for each specimen type. The pH of the hardened cement paste was 12.8, as measured according to ASTM D4972 [40].

2.3. Conditioning

Specimen conditioning consisted in immersing the bare GFRP bars and the concrete-embedded GFRP bars in three different types of solutions, i.e., tap water (TW), saline solution (SS), and alkaline solution (AS) for 28, 56, and 112 days (672, 1344, and 2688 h, respectively) at three different temperatures (23 °C, 60 °C, and 80 °C), as shown in Fig. 2. These conditioning temperatures were selected because they are well below the glass transition temperature of the GFRP bars of 125.8 °C, as reported in Table 1. Bank et al. [41] suggested these temperature levels accelerate the degradation effect of aging but ensure that the GFRP bars will not undergo a change in degradation mechanism. In addition, the GFRP bars were lightly loaded or stressed in a number of reported GFRP-reinforced concrete structures and without concrete cracking [24,42]. Thus, the effect of stress was not considered in this study. Tap water was used to simulate the high-moisture condition, while the saline solution was prepared in accordance with ASTM D 1141-98 [43] using 3.5% NaCl solution by weight to match the salinity of saltwater in a coastal environment. The alkaline solution—prepared in accordance with ASTM D7705 [44] —had a pH of 12.7 to simulate a concrete environment. During the conditioning, the glass containers were covered to minimize solution evaporation. They were also constantly checked and refilled as needed to ensure the specimens remained completely submerged. The glass containers were placed inside a programmable oven for the conditioning at elevated temperatures of 60 °C and 80 °C.

Table 1
Summary of the test methods and number of specimens.

Properties	Test Method	No. of Specimens	Average	Standard Deviation
Physical properties				
Cross-sectional area (mm^2)	CSA-S806, Annex A [32]	9	83.8	1.9
Fiber content by weight (%)	ASTM D3171-15 [33]	9	80.9	0.2
Transverse CTE ($x10^{-6}$/°C)	ASTM E1131-08 [34]	9	20.7	2.3
Void content (%)	ASTM D5117-09 [35]	9	0	0
Water absorption at 24 h (%)	ASTM D570-98 [36]	15	0.15	0.01
Water absorption at saturation (%)	ASTM D570-98 [36]	15	0.19	0.01
Cure ratio (%)	ASTM E 1356-08 [37]	15	100	0
T_g (°C)	ASTM E 1356-08 [37]	15	125.8	1.3
Mechanical properties				
Interlaminar shear strength, S_u (MPa)	ASTM D4475-02 [38]	6	54.7	1.1

2.4. Moisture uptake

The percentage moisture uptake of the conditioned GFRP bars (only those samples not embedded in concrete) were measured every day during the first 28 days and every week thereafter according to ASTM D570 [36]. The samples were weighed to determine the initial mass (m_0) prior to conditioning. To measure the moisture uptake, the specimens were removed from the solution, quickly washed in tap water, dried with paper towels, and then immediately weighed (m_{1B}). Lastly, the specimens were post-conditioned and reweighed (m_{1A}). The percentage moisture uptake is expressed as the ratio of the weight of the water absorbed by the bars to the weight of the dry bars (*weight of wet specimen/weight of dry specimen*). It was calculated with Eq. (1):

$$Mass\ gain\ (\%) = \frac{m_{1A} - m_0}{m_0} \times 100 \tag{1}$$

2.5. Short-beam shear test

The long-term performance of the GFRP bars was evaluated using their interlaminar shear strength (ILSS) in accordance with ASTM D4475-02 [38]. The embedded GFRP bars were carefully extracted with a small flat-head hammer to prevent damage. It is important to note that there were no cracks in the concrete cover after conditioning, indicating no bar expansion and no solution ingress through cracks. The short-beam shear test was performed on GFRP bars with a clear span of $3d_b$ and overhang length of $0.5d_b$. Micelli and Nanni [16] indicated that this span would prevent flexural effects that could change the desired shear failure mode of the bars. Six samples from each group were tested to examine the interlaminar shear strength, S_u, of the conditioned and unconditioned GFRP bars. The interlaminar shear testing was performed with an MTS 100 kN testing machine with a controlled displacement rate of 1.3 mm/min. Fig. 3a shows the actual test setup. The interlaminar shear strength was calculated as $S_u = 0.849P/d_b^2$, where P is the shear failure load (N) and d_b is the bar diameter (mm). Table 4 presents the average and standard deviation of the ILSS for all the tested specimens.

The specimens were designated as being embedded in concrete (CE) or not, followed by the type of conditioning environment (TW for tap water, SS for saline solution, and AS for alkaline solution), followed by the conditioning temperature (RT for room temperature or 23 °C, 60 for 60 °C, and 80 for 80 °C) and exposure duration (28, 56, or 112 days). For example, specimen CE-TW-60-112 is a GFRP bar embedded in concrete and conditioned in tap water at 60 °C for 112 days.

3. Results and discussion

3.1. Moisture uptake

Moisture uptake is an important factor that can significantly affect the mechanical and durability properties of GFRP bars [11]. Micelli and Nanni [16] pointed to strong evidence that the rate of degradation of GFRP bars exposed to fluid environments is related to the rate of fluid sorption. GFRP bars embedded in reinforced-concrete elements can absorb moisture and water, which penetrate the resin and affect the fiber–resin interface. Moreover, the water volume expansion at low temperatures degrades the polymeric resin, thereby reducing the shear strength of GFRP bars [14].

In this study, the percentage moisture uptake at different condi-

a. GFRP bars b. GFRP bars embedded in concrete

Fig. 1. Specimen types.

Fig. 2. Conditioning of GFRP bars.

tioning times is used to obtain quantitative information about the diffusion properties of GFRP bars in different solutions. Fig. 4 shows the plot of the moisture uptake at saturation of bare GFRP bars conditioned in tap water (TW), alkaline solution (AS), and saline solution (SS) and at different temperatures (23 °C, 60 °C, and 80 °C) against the square root of the exposure time in seconds. Most or the moisture uptake was expected to occur at the bar surface rather than at the ends, since the area of the bar surface is at least 8 times greater than that of the bar ends. On the other hand, Table 2 summarizes the peak moisture uptake of the GFRP bars (M_m) and the apparent diffusivity (D) in the different solutions and at the different conditioning temperatures as calculated with Fick's equation (Eq. (2)) for a bar-shaped specimen, as presented by Aiello et al. [26]:

$$D = \pi\left(\frac{d_b}{4.M_m}\right)^2\left(\frac{M_2 - M_1}{\sqrt{t_2} - \sqrt{t_1}}\right)^2 \quad (2)$$

where d_b is the nominal bar diameter and M_1 and M_2 are the moisture contents at times t_1 and t_2, respectively. Both t_1 and t_2 are taken when the change in moisture uptake was relatively linear, varying with the square root of time as suggested by Karbhari and Zhang [9].

There is an obvious amount of scatter in the measured percentage moisture uptake, primarily due to the relatively small mass change. In general, the moisture uptake process of GFRP bars followed Fick's second law of diffusion. The moisture uptake increased almost linearly in the first 10 days ($s^{1/2}$ of around 1000) and tended to stabilize after that with the increasing square root of exposure time due to the bars having already reached saturation. This two-stage moisture uptake is typical of high volume-fraction unidirectional composites, as also suggested by Karbhari and Xian [45]. Of particular note is the difference in percentage moisture uptake of the GFRP bars depending on the solution type and conditioning temperature. The percentage moisture uptake was higher at higher exposure temperatures, indicating that temperature has a direct effect on water absorption. As with the percentage moisture uptake, apparent diffusivity values also increased with the temperature. In comparing specimens with similar temperature exposures, the GFRP bars conditioned in the alkaline solution had the highest moisture uptake (0.16%–0.22%), as shown in Fig. 4c followed by the bars conditioned in tap water (0.14%–0.18%) and saline solution (0.12%–0.15%), as shown in Figs. 4a and b, respectively. The high moisture uptake in AS is due to the free OH ions in the solution, which can break the Si–O–Si structures and promote diffusion of more water within the bar, as found by Chen et al. [46]. On the other hand, Bank et al. [41] indicated that composite take up more moisture in water than in saline solution because salt ions are larger than water ions. Overall, it is promising that the moisture uptake of the bars was very low and similar to that of the percentage absorption of the straight GFRP bar samples (0.14%)

a. Test setup b. Failure

Fig. 3. Short-beam shear test.

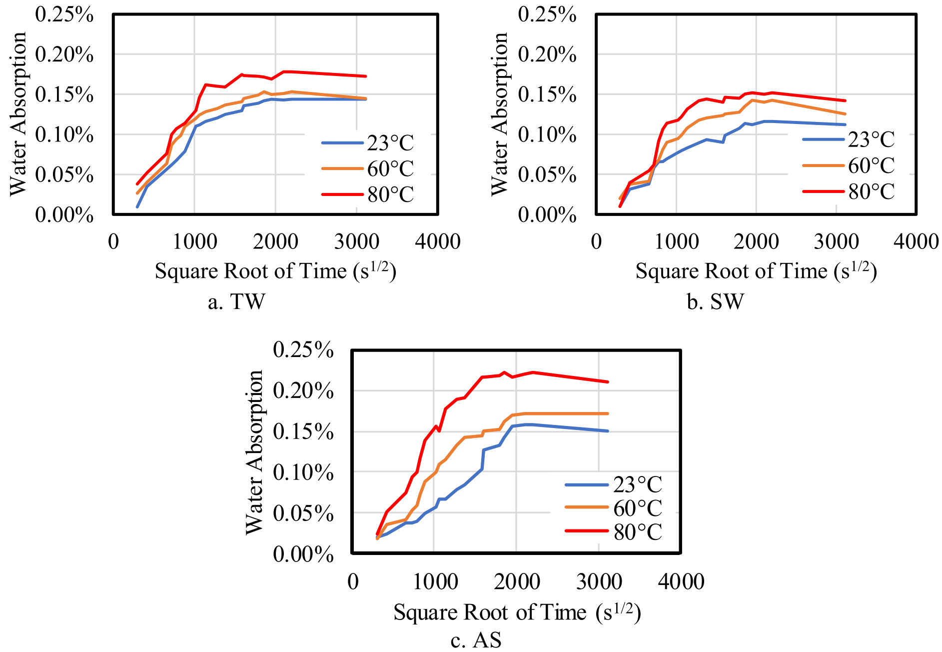


Fig. 4. Moisture absorption process of GFRP bars conditioned in different solutions and at different temperatures.

Table 2
Comparison of the peak moisture absorption (M_m) and apparent diffusivity (D).

Temperature (°C)	Tap Water		Saline Solution		Alkaline Solution	
	M_m (%)	D (x10^{-7} mm^2/s)	M_m (%)	D (x10^{-7} mm^2/s)	M_m (%)	D (x10^{-7} mm^2/s)
23	0.14	1.01	0.12	0.71	0.16	1.75
60	0.15	1.67	0.14	1.32	0.17	2.13
80	0.18	2.55	0.15	1.92	0.22	3.14

measured by Benmokrane et al. [24]. Moreover, the measured apparent diffusivity—0.71 × 10^{-7} mm^2/s—is in good agreement with the values reported by Karbhari et al. [9] for E-glass/vinyl-ester composites exposed to concrete-based alkaline solutions. The low moisture uptake and apparent diffusivity values were also due to the vinyl-ester resin having fewer hydrophilic properties since it contains fewer polar ester moieties, as found by Chin et al. [7]. This behavior also suggests that the moisture was absorbed only through the resin-rich surface of the GFRP bars, as suggested by Mouzakis et al. [47]. The effect of this absorbed solution on the fiber–matrix interface was evaluated with the interlaminar shear strength test and discussed in the next section.

3.2. Interlaminar shear strength (ILSS)

The long-term properties of the bare and embedded GFRP bars in different solutions was evaluated according to their interlaminar shear strength (ILSS). Gooranorimi and Nanni [17] and Benmokrane et al. [24] used this property as a useful parameter for durability evaluation, especially with GFRP bars under in-field conditions, since bars that can be extracted from the concrete core are mostly of limited length.

All the specimens failed due to horizontal shear originating at the edge of the bars and developed along bar length. A minor difference observed is the location of the interlaminar shear crack along the bar cross section. In the control and the GFRP bars conditioned for 28 and 56 days, the crack occurred mostly at bar mid-depth, compared to below mid-depth in the bars conditioned for 112 days, as shown in Fig. 3b, indicating a weaker fiber–matrix interface in this location. This crack in the lower portion of the conditioned bars could be due to the outermost resin-rich surface of the bars being already saturated with solution. While the highest shear stress was induced in the middle of the cross section, the outer portion of the bar is weaker and more prone to shear failure. This would cause degradation, creating weak zones along which transverse cracking could easily occur during testing.

Tables 3 and 4 summarize the average and standard deviation of the ILSS of the bare and concrete-embedded GFRP bars exposed to different solutions and at different temperatures, respectively. The low values of the standard deviation (maximum of 8.3%) suggest adequate bar extraction, since Gooranorimi and Nanni [17] and Benmokrane et al. [24] stated that this procedure is a major factor that could influence the ILSS of GFRP bars. The average ILSS of the unconditioned bars was 54.9 MPa with a standard deviation of 0.7 MPa. This is comparable to the properties reported by Benmokrane et al. [1] (see Table 1) as these bars were obtained from the same production lot.

Generally, the ILSS of the GFRP bars (see Tables 3 and 4) decreased as the exposure temperature and duration increased under all exposure conditions. This can be attributed to the increase in percentage moisture uptake of the bars with exposure duration, which leads to degradation of the fiber–matrix interface. When the bars absorbed moisture, the resin-rich surface degraded. The bond between the matrix and fibers located on the outer part of the bar gradually decreased and the bar resistance started to decrease. The GFRP bars had low water absorption (maximum of 0.22% for the AS-conditioned bars after 112 days), indicating that only a very thin layer was affected by the solution. Softening of the resin at the edge of the bars was very critical in the short-beam shear test, as this location is subjected to a high level of shear stress. Nkurunziza et al. [12] highlighted that the interface between the glass fibers and resin controls GFRP-bar resistance to different environments. In our study, however, the ILSS of the concrete-embedded GFRP bars was generally higher than that of the bare GFRP bars under similar immersion conditions. This behavior can be attributed to the limited availability of moisture around the bars and the lower temperature condition for the concrete-embedded bars. This means that the concrete cover can protect the bars from absorbing moisture from the environment that the bars is exposed to. On the contrary, Ceroni et al. [15] indicated that the cut ends directly expose the fibers to the external environment, thereby degrading bar durability of bare GFRP bars. Benmokrane et al. [1] however, found that the moisture uptake of FRP bars is a function of the exposed bar surface. Thus, most of the moisture uptake would occur on

Table 3
Interlaminar shear strength in MPa of bare GFRP bars.

Temp (°C)	Tap Water			Saline Solution			Alkaline Solution		
	28 days	56 days	112 days	28 days	56 days	112 days	28 days	56 days	112 days
23	47.4 (0.1)	37.2 (0.7)	24.3 (0.7)	51.9 (0.8)	44.6 (0.5)	37.6 (0.4)	44.6 (0.2)	32.1 (0.2)	23.2 (0.8)
60	41.7 (0.3)	31.5 (0.7)	18.4 (1.2)	49.3 (1.6)	39.1 (0.4)	29.8 (0.6)	34.5 (0.5)	27.4 (0.4)	16.6 (0.7)
80	37.4 (0.7)	26.6 (0.6)	9.3 (0.8)	46.6 (0.3)	35.3 (0.7)	23.7 (0.6)	28.7 (2.4)	20.9 (0.9)	12.9 (0.9)

The numbers inside parentheses are the standard deviations.

Table 4
Interlaminar shear strength in MPa of concrete-embedded GFRP bars.

Temp (°C)	Tap Water			Saline Solution			Alkaline Solution		
	28 days	56 days	112 days	28 days	56 days	112 days	28 days	56 days	112 days
23	50.3 (2.0)	49.3 (0.8)	48.1 (0.3)	52.5 (1.3)	50.8 (0.8)	49.7 (1.9)	47.5 (0.2)	45.9 (0.4)	44.1 (0.5)
60	48.8 (0.6)	47.0 (0.6)	45.6 (0.9)	50.8 (1.0)	49.4 (0.5)	47.7 (0.7)	43.4 (0.2)	41.5 (0.7)	40.3 (1.0)
80	43.5 (0.6)	41.5 (0.3)	38.5 (0.4)	47.3 (0.9)	46.1 (0.2)	44.6 (0.2)	41.3 (1.8)	39.1 (0.8)	36.3 (0.4)

The numbers inside ( ) are the standard deviation.

the bar's surface than at the ends. The increased temperature further increased water absorption, resulting in faster degradation of the fiber–matrix interface, leading to the decrease in the ILSS.

3.3. ILSS retention

The durability performance of the GFRP bars was appraised based on ILSS retention after conditioning in different solutions and for different exposure durations. The ILSS retention was calculated by dividing the average ILSS for each bar (see Tables 4 and 5) to that of the average property value for the reference specimen. Fig. 5 shows the ILSS retention values of the conditioned bare GFRP bars, while Fig. 6 shows the ILSS retention values for the concrete-embedded GFRP bars. The following important information can be drawn from Fig. 5.

- The AS was the most aggressive solution, affecting GFRP-bar ILSS, because the bars had the highest absorption in this solution. In the case of TW immersion, the ILSS retention values of the bare GFRP bars at the end of 28 days of immersion at 23 °C, 60 °C, and 80 °C were 86%, 76%, and 68% in TW; 94%, 90%, and 85% in SS; and 81%, 63%, and 52% in AS.
- At the end of 56 days of immersion at 23 °C, 60 °C, and 80 °C, the retention levels were 68%, 57%, and 48% in TW; 81%, 71%, and 64% in SS; and 58%, 47%, and 38% in AS.
- At the end of 112 days of immersion at 23 °C, 60 °C, and 80 °C, the retention levels were 48%, 41%, and 29% in TW; 69%, 54%, and 43% in SS; and 38%, 30%, and 23% in AS. These results show that the ILSS of the GFRP bars was affected by immersion time and level of temperature exposure.

Fig. 6 suggests the following with respect to the concrete-embedded GFRP bars:

- At the end of 28 days of immersion at 23 °C, 60 °C, and 80 °C, the retention values for the GFRP bars were 92%, 89%, and 79% in TW; 96%, 93% and 86% in SS; and 87%, 79%, and 75% in AS.
- At the end of 56 days of immersion at 23 °C, 60 °C, and 80 °C, the retention levels were 90%, 86%, and 76% in TW; 93%, 90%, and 84% in SS; and 84%, 76%, and 71% in AS.
- At the end of 112-day immersion at 23 °C, 60 °C, and 80 °C, the retention levels were 88%, 83%, and 70% in TW; 91%, 87%, and 81% in SS; and 80%, 73%, and 68%in AS. These results clearly show that direct immersion in alkaline solutions deteriorates GFRP bars more severely than does exposure to concrete alkali.

The experimental results presented above clearly indicate that high temperatures had an immediate impact on ILSS degradation. The ILSS retention decreased as the exposure temperature and duration increased. The significant ILSS degradation could be explained by matrix degradation at higher temperatures, which, in turn, could have affected the fiber–matrix interface. Water molecules penetrating through the fiber–resin interface have a plasticization effect and cause hydrolysis reactions leading to interfacial fracture. Moreover, the AS was the most aggressive of the three solutions with respect to ILSS. Chen et al. [46] suggested that the bond in the vinyl-ester matrix is highly prone to degradation in the presence of the free OH ions in the alkaline solution. The process involves hydrolysis reactions in the ester group, leading to leaching in the matrix and damage to the glass fibers, which weakens the interfacial-bond strength. Furthermore, Micelli and Nanni [16] indicated that the high pH of the pore water solution during concrete hydration might subject the glass fibers to chemical attack. This agrees with the findings of Nkurunziza et al. [12], namely that the most critical degradation of GFRP bars occurs in alkaline solutions. Wang et al. [29] had similar findings, measuring higher rates of ILSS degradation in higher alkali-ion solutions. Their measured reduction in ILSS was significantly lower than the almost 90% reduction measured by Micelli and Nanni [16] for GFRP bars immersed in alkaline solution for 42 days at 60 °C. In contrast, Ashrafi et al. [27] measured a maximum reduction of 15% in the ILSS of E-glass FRP bars subjected to water-vapor condensation after 3000 h. This conditioning only affected the very thin resin rich surface of the bars. Similarly, Benmokrane et al. [1] measured a 12% decrease in ILSS for 9.5 mm diameter GFRP bars conditioned in an alkaline solution for 90 days at 60 °C. It should be noted, however, that had the alkaline solution used by Benmokrane et al. [1] had a pH of only 12.6, compared to 12.7 in our study. This may explain the slightly higher ILSS reduction of the GFRP bars investigated in this work compared to that of Benmokrane et al. [1].

Table 5
$\frac{E_a}{R}$ and *TSF* values for all conditioning cases.

Bar	Accelerating Agent	$\frac{E_a}{R}$	23 °C	30 °C	60 °C	80 °C
Bare GFRP bars	**Time Shift Factor for bare GFRP bars (TSF_{bb})**					
	TW	1217	1.00	1.10	1.58	1.94
	SS	1240	1.00	1.10	1.59	1.97
	AS	2250	1.00	1.19	2.33	3.41
Embedded GFRP bars	**Time Shift Factor for embedded GFRP bars in concrete (TSF_{cb})**					
	TW	8946	1.00	2.01	28.64	131.06
	SS	4599	1.00	1.43	5.61	12.26
	AS	4642	1.00	1.44	5.70	12.55

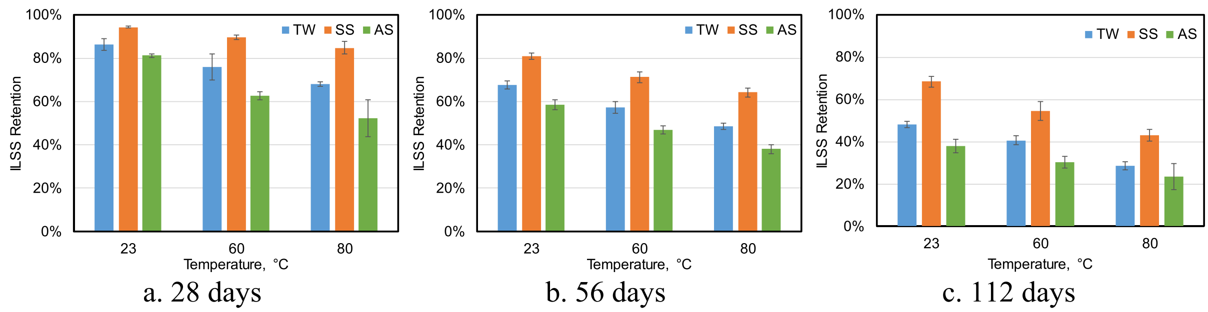


Fig. 5. Interlaminar shear-strength retention of bare GFRP bars.

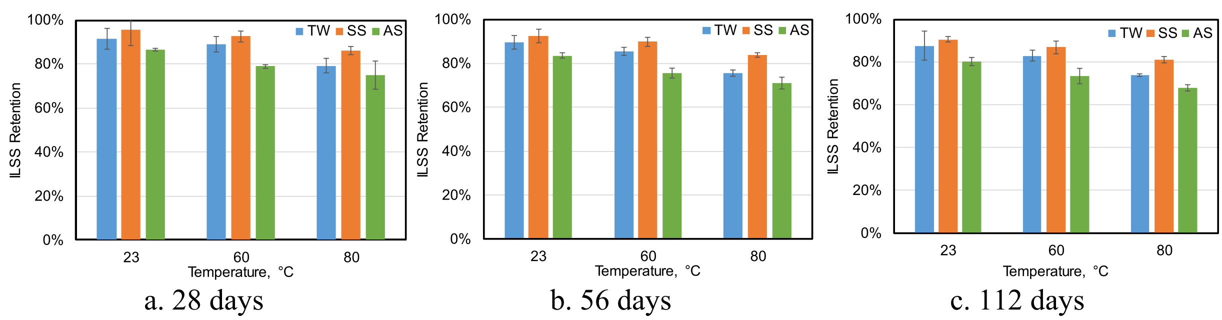


Fig. 6. Interlaminar shear-strength retention of concrete-embedded GFRP bars.

The decreased retention of ILSS in the GFRP bars exposed to TW and SS is in agreement with the results of the research studies presented in D'Antino et al. [25] in which vinyl-ester-based GFRP bars were embedded in concrete and conditioned in tap water and saline solution at 50°C-60 °C. Their experienced a significant decrease in strength as the exposure time increased. Wang et al. [29] measured only a 50% retention in the ILSS of GFRP bars exposed to seawater solution at 55 °C for 84 days. Other researchers [22,48] have indicated that tap water and saltwater affect GFRP-bar durability similarly. The results of our study suggest a higher decrease in ILSS exposed to tap water than to seawater. D'Antino et al. [25] reported a similar observation: GFRP bars conditioned in a saline solution experienced less degradation than those conditioned in plain water at the same temperature. This could be due to the higher moisture uptake of the GFRP bars in TW than in SS, as shown in Fig. 3. This could be also explained by the large salt molecules in the saline solution slowing the diffusion of water molecules by blocking the paths by which water diffuses into the bars. Mouzakis et al. [47] had similar findings. Almusallam et al. [21] further indicated that the lower water absorption of GFRP bars in SS than TW can be attributed to a very thin layer of salt forming on the specimen surface—especially at higher temperatures—which decreases the diffusion rate of the solution into the GFRP bars. In all exposure temperatures considered in our study, the rate of diffusion of GFRP bars exposed in SS was lower than that in TW.

For the concrete-embedded GFRP bars herein, the ILSS dropped gradually over time. In contrast, a significant decrease occurred in the GFRP bars, especially those in AS, even after only 28 days. For example, the bare GFRP bars retained only 63% of their ILSS, while the concrete-embedded bars retained 79% at 60 °C in AS. The 21% decrease in ILSS for the concrete-embedded bars is almost the same as the 23% decrease in ILSS measured by Gooranorimi and Nanni [17] for 15.9 mm diameter GFRP bars extracted from a concrete bridge deck after 15 years of service. It is important, however, to note that these researchers only tested one extracted sample, which is insufficient to prove the reliability of the test results. Similarly, Micelli and Nanni [16] estimated that GFRP bars subjected to alkaline exposure for 21 days at 60 °C would correspond to 14 years in an actual concrete structure. The higher ILSS retention of the concrete-embedded GFRP bars was due to the cement, which may not have been completely saturated, and to the moisture on the outside of the concrete not being in direct contact with the bars. The concrete cover minimized the water absorption of the GFRP bars, whereas the bare GFRP bars were completely immersed in solution that wet the entire external surfaces of the bars. As a result, the decomposition reaction of the GFRP bars embedded in the concrete would be slower because of the lack of an oxidation reaction in the polymer matrix.

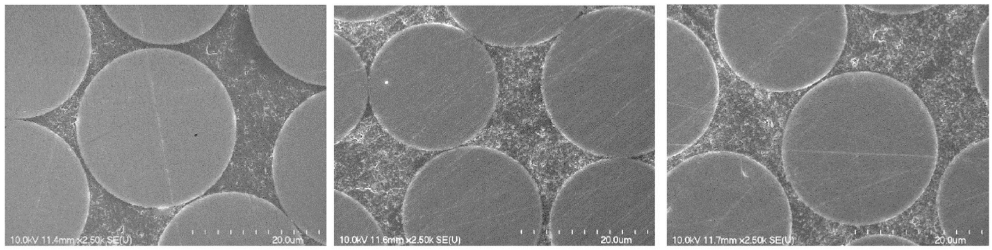
4. SEM and FTIR observations

Scanning-electron-microscopy (SEM) observations were performed to assess the microstructure of the GFRP bars before and after conditioning after 112 days using the JEOL JSM-840A SEM (JEOL, Akishima, Tokyo, Japan). All of the specimens observed under SEM were cut, polished, and coated with a thin layer of gold–palladium using a vapor-deposit process. SEM observations were performed on both the bar cross section and at the fiber–matrix interface. Similarly, Fourier transform infrared spectroscopy (FTIR) was performed to study the changes in the chemical composition of the matrix at the bar surface. These observations were to determine the potential degradation of the polymer matrix, glass fibers, or interface, as applicable, due to solution penetration. The aim was to link these observations to possible changes in ILSS and bar chemical composition after conditioning.

4.1. SEM

Fig. 7 provides micrographs of bar cross sections at 2500 times magnification. There were no observable differences between the bare bars and the concrete-embedded bars exposed to different solutions. This could be due to the bars not being under the stress, so the fiber–matrix interface remained intact. Moreover, no pores or air bubbles were observed, indicating the high quality of the manufacturing process.

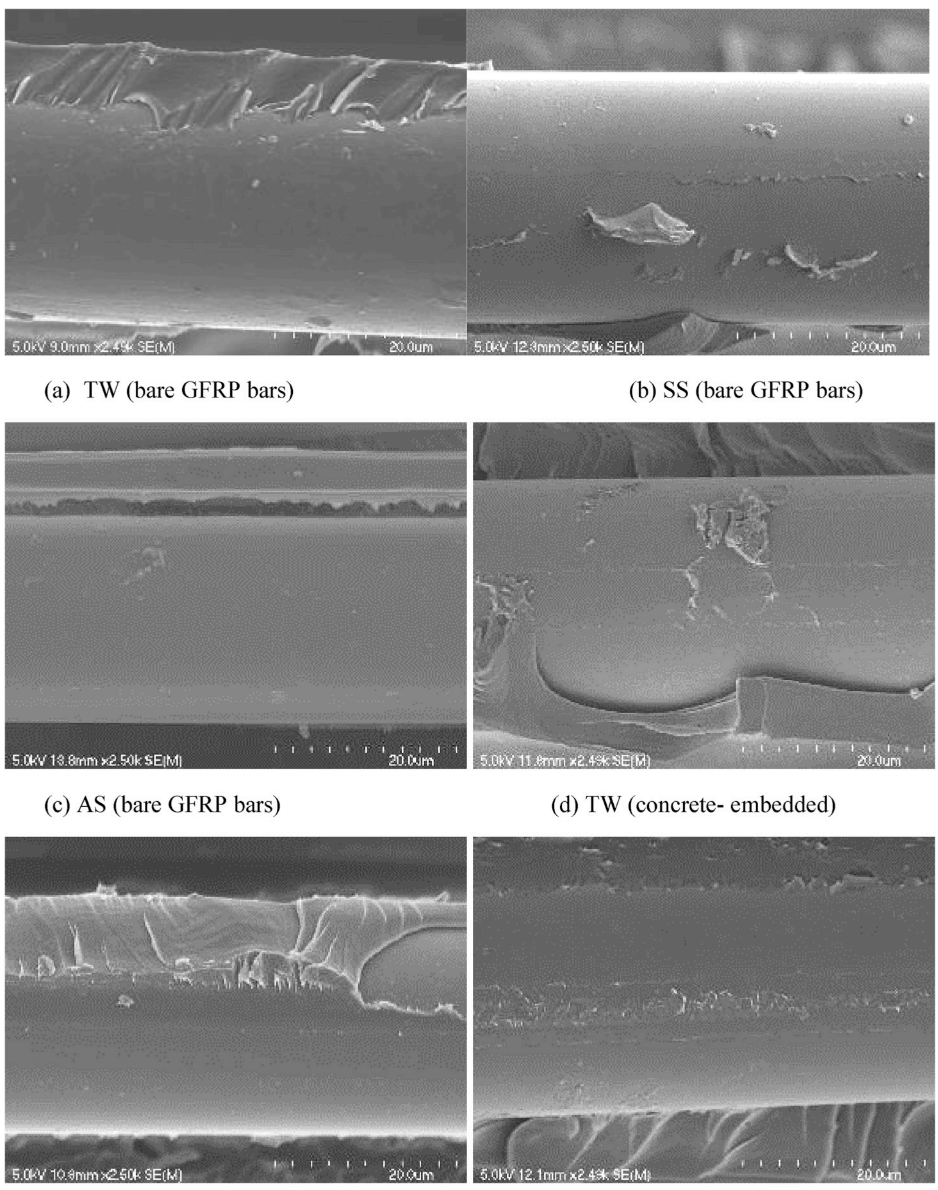
After the short-beam testing, SEM was also performed on the fracture zones near the ends of the GFRP bars conditioned for 112 days to investigate the mechanisms of failure at the fiber–matrix interface, as shown in Fig. 8. The SEM shows that the surface fracture was dominated by matrix fracture, since the surface matrix layer covers and protects the



(a) TW (b) SS (c) AS

Fig. 7. SEM observations of bar cross sections.

glass fibers was lost in some areas. None of the specimens exhibited fiber damage within the bar. The bare GFRP bars exposed to TW and SS (Figs. 8a and b, respectively) had resin adhering to fibers, representing a good interface. In contrast, there was less resin on the fiber surfaces of the bars exposed to AS, indicating decreased bonding strength between the fibers and resin. Fig. 8c shows some smooth fibers with almost no resin residue, which indicates that the rupture occurred partly at the interface. In contrast, the fiber surfaces of the concrete-embedded GFRP bars had more resin coverage than the bare GFRP bars, but still could have lost a certain degree of adhesion. This observation explains the



(a) TW (bare GFRP bars) (b) SS (bare GFRP bars)

(c) AS (bare GFRP bars) (d) TW (concrete- embedded)

(e) SS (concrete- embedded) (f) AS (concrete- embedded)

Fig. 8. SEM observations at the failure surface of conditioned GFRP bars.

lower ILSS retention of the bare GFRP bars compared to that of the concrete-embedded bars after conditioning. Moreover, a lot of residual resin on the fiber surfaces was observed in the case of the concrete-embedded bars exposed to TW and AS (Figs. 8d and e, respectively). This suggests better bonding between the fiber and vinyl-ester resin than in the bars exposed to AS (Fig. 8f). This observation also suggests that the integrity of the fiber–matrix interface can be better evaluated by observing the fracture surface than the ends of the GFRP bars.

The damage to the fiber–resin interface and ILSS decrease were due to fluid penetration, resulting in moderate moisture content after immersion. The moisture absorbed by the bars, combined with the exposure temperature, induced stress in the material with consequent interfacial damage and decreasing the ILSS with time. Mouzakis et al. [47] indicated that absorbed water can disrupt the interfacial bonds between the fibers and matrix. Moreover, Ceroni et al. [15] indicated that deterioration of this interface reduces the load-transfer capacity between fibers, thereby decreasing mechanical properties. Davalos et al. [20] highlighted that the integrity of the fiber–matrix interface is critical for load transfer between fibers, and that interfacial degradation weakens the composite materials. Moreover, Nkurunziza et al. [12] pointed out that the chemical bond between the coupling agent and the surface of the glass fibers is not stable in the presence of moisture and alkalis. Bar absorption of moisture and alkalis gradually destroyed the bond, damaging the interface and reducing the stress transfer efficiency between the fibers and matrix. This presence of water, combined with high pH levels, considerably affects the physical and chemical degradation at the fiber–matrix interface.

4.2. FTIR

FTIR spectra of the GFRP bars were recorded with a Nicolet Magma 550 spectrometer (see Fig. 9). For both the bare (Fig. 9a) and concrete-embedded (Fig. 9b) bars, the FTIR spectra focused on OH units at around 3500 cm^{-1} and CH groups at around 2900 cm^{-1}. When degradation by hydrolysis occurred, the OH peak dramatically increased compared to the CH peak, which remained constant. The hydroxyl peak did not show any significant changes, which indicates that no significant hydrolysis of the GFRP bars occurred, except for the relatively higher intensity of the O–H stretching band at 3400 cm^{-1} for the bare GFRP bars exposed to AS. Chin et al. [7] suggested that this spectral change is consistent with ester hydrolysis, in which ester functional groups are converted to hydroxyl and carboxylic acid products. However, this higher intensity was only observed at the bar surface indicating that the water absorption was only concentrated in the thin resin rich area of the GFRP bars. This indicates that ageing might have occurred on the surface of the GFRP bars. However, it is also important to note that the OH stretching might not be due to hydrolysis. Vinyl ester naturally contains OH, and if unevaporated water or alkalis are present in the bars, the amount of OH will also increase. This degradation in materials accounts for the fractured fiber–matrix interface observed under SEM and the significant loss in ILSS of the bare GFRP bars exposed to AS.

5. Prediction of the long-term behavior and service life of GFRP-Bar ILSS

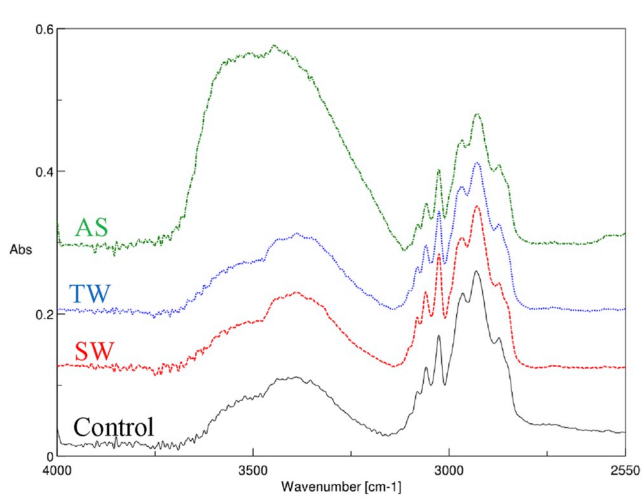
Aiello et al. [26] stated that reliably predicting the long-term behavior of civil infrastructure subject to the action of environmental factors is a complex problem. It requires accelerated aging through hygrothermal exposure for the long-term assessment of materials durability and relies on the superposition of temperature and moisture to enhance and speed up environmental degradation. Moreover, Davalos et al. [20] highlighted that only a few studies focused on the development of life-cycle durability prediction models for FRP bars in concrete environments. Naya et al. [49] suggested that the most accurate method is the Arrhenius method for materials exposed to temperatures below their glass transition temperatures. Thus, the long-term ILSS performance of GFRP bars investigated in this study was predicted in accordance with the Arrhenius relation and the procedure implemented by Bank et al. [41]. As a requirement, at least three elevated temperatures and three exposure durations are necessary to perform an accurate prediction based on the Arrhenius law [48], which was conducted in this study.

5.1. Arrhenius model

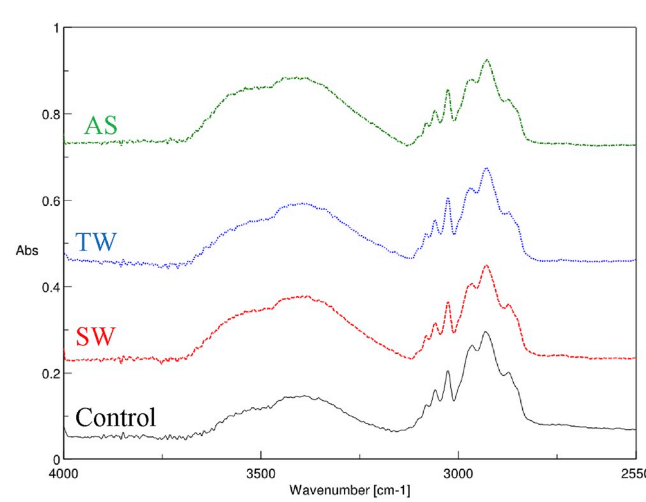
The Arrhenius equation is used to express the degradation rate for materials with time as according to Nelson [50] with Eq. (3):

$$k = A \times e^{\left(\frac{-E_a}{RT}\right)} \tag{3}$$

where k is the degradation rate (1/*time*), A is a constant based on material properties, E_a is the activation energy, R is the universal gas constant, and T is the temperature in Kelvin. The basic assumption in this prediction model is that the material properties of the GFRP bars are not affected by temperature during exposure and the rate of degradation



(a) Bare bars



(b) concrete-embedded bars

Fig. 9. FTIR of GFRP bars.

is accelerated with the increase in temperature. Therefore, analyzing procedures to identify the factors in the rate of degradation consist in transforming Eq. (3) to produce a linear equation between the time ($t = 1/k$) and the inverse of temperature ($1/T$) by taking the natural logarithm of both sides of the equation, as shown in Eq. (4):

$$\ln\left(\frac{1}{k}\right) = \frac{-E_a}{R}\left(\frac{1}{T}\right) + \ln(A) \tag{4}$$

Afterwards, the Arrhenius method is carried out by plotting the natural logarithm of time needed to achieve 90%, 80%, 70%, and 60% strength retention (*SR*) of the ILSS of the GFRP bars with the inverse temperature ($1000/T$) in Kelvin to obtain the regression coefficient ($\frac{E_a}{R}$) value. The time to reach various levels of property retention at different aging temperatures was calculated by substituting different values of ILSS strength retention in the regression equations in Eq. (4). This value was expressed as the slope of the linear equations, as can be seen in Fig. 10 for the bare bars and Fig. 11 for the concrete-embedded bars immersed in different solutions. Regression analyses were performed to determine the line of best fit, as shown in these figures. The straight lines were nearly parallel to each other, indicating that the accelerated aging tests were valid, and this model could be applied to describe the ILSS degradation of GFRP bars. Moreover, the R^2 of the regression line is at least 0.92, which is well above 0.80, as indicated by Benmokrane et al. [48]. The average of these slopes represents the $\frac{E_a}{R}$ values.

The regression coefficient ($\frac{E_a}{R}$) was then used to determine the relative time-shift factor (*TSF*) for all conditioning cases for two different temperatures (See Eq. (5)), as suggested by Dejke [51]. In Eq. (5), t_1 and t_2 (days) are the times required to reach a certain strength retention (*SR*); k_1 and k_2 are the degradation rates corresponding to t_1 and t_2, respectively; T_{ref} and T_0 are the reference and exposure temperatures (Kelvin), respectively. The main purpose of the *TSF* is to transform the time taken in accelerated tests for a known exposure condition and temperature and correlate this with the actual service life of GFRP bars in concrete. In our study, the reference temperature was 23 °C. Table 5 provides the $\frac{E_a}{R}$ and *TSF* values for all conditioning cases. Table 5 shows that the $\frac{E_a}{R}$ values for the concrete-embedded bars were much higher than those for the bare bars, indicating that the embedded bars required higher activation energy to cause bar degradation. This also means that the embedded bars had a lower degradation rate than the bare bars, which were directly immersed in the solutions. This indicates that submersion in water and other solutions, as opposed to contact with concrete, intensifies degradation of GFRP bars.

$$TSF = \frac{t_1}{t_2} = \frac{k_2}{k_1} = \frac{A \times e^{\left(\frac{-E_a}{RT_0}\right)}}{A \times e^{\left(\frac{-E_a}{RT_{ref}}\right)}} = e^{\left(-\frac{E_a}{R}\right)\left(\frac{1}{T_0} - \frac{1}{T_{ref}}\right)} \tag{5}$$

5.2. Predicting the long-term behavior and constructing master curves

A number of methods have been suggested for predicting the long-term behavior of GFRP bars after accelerated testing [41,51]. Chen et al. [46] and Ali et al. [52,53], however, suggested that developing a master curve containing a plot for all the data used for analysis would yield more precise and accurate predictions. This master curve consists of the time required to reach a specific *SR* corresponding to its *TSF*, considering the effect of temperature. Accordingly, Figs. 12 and 13 show the master curves at a temperature of 23 °C for the bare and embedded bars under the conditioning environments used in our study. These curves show the SR in percentage along the y-axis against time *t* in days along the x-axis. It can be observed that master curves can be expressed by a logarithmic equation with an R^2 of at least 0.94. This means that the *SR* versus time model proposed by Bank et al. [41] is valid and it yielded good predictions using Eq. (6) in its general form, where *a* and *b* are regression constants.

$$SR = a\ln(t) + b \tag{6}$$

5.3. Equivalent service life for GFRP bars in concrete environments

The strength retention measured for the bare GFRP bars exposed to the different solutions at different temperatures and for different exposure times was correlated to the strength retention of the concrete-embedded bars to determine the equivalent service life (t_{ESL}) for GFRP bars in concrete environments. This approach is proposed based on the *TSF* values (Eq. (5) and Table 5) and the master curves in Figs. 12 and 13 wherein the strength retention of the bare GFRP bars were correlated to that of the GFRP bars in actual concrete structures at any condition. This prediction model might only be applicable to GFRP bars subjected to low stress levels. For example, predicting the equivalent service life of GFRP

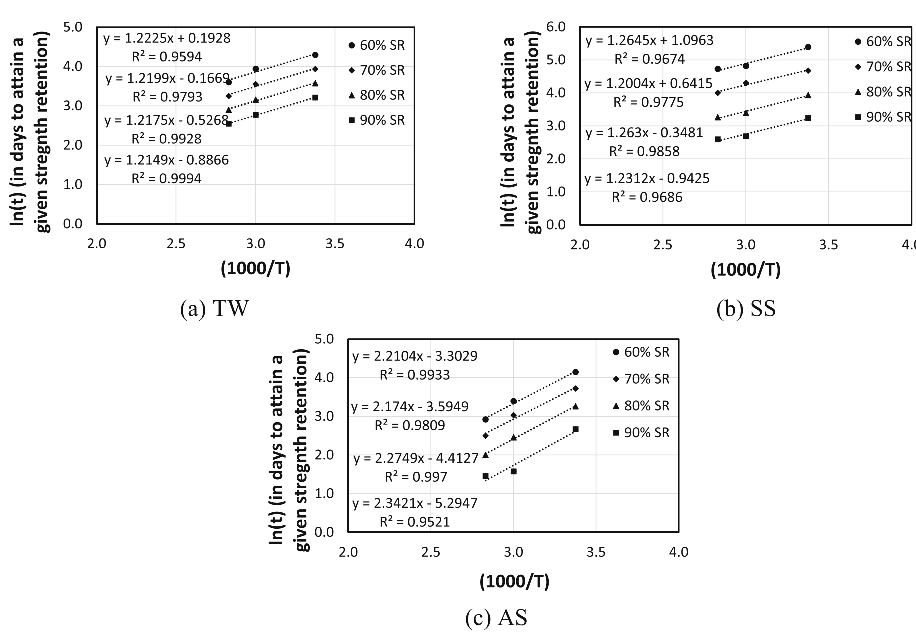


Fig. 10. Arrhenius plots for the service life of the bare GFRP bars.

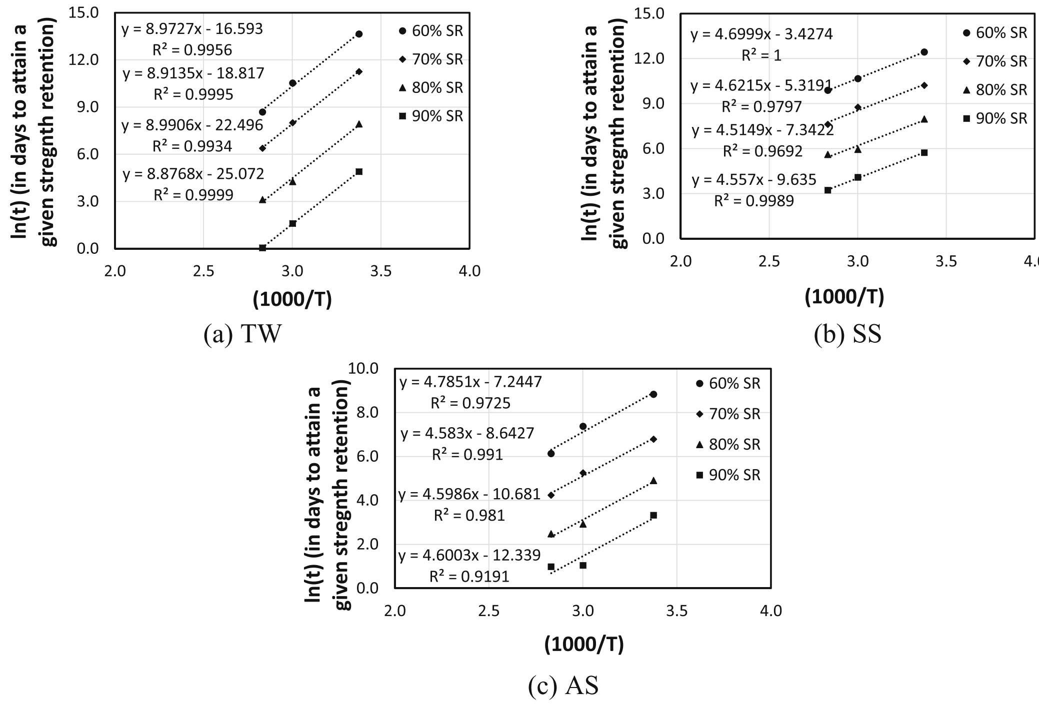


Fig. 11. Arrhenius plots for the service life of the concrete-embedded GFRP bars.

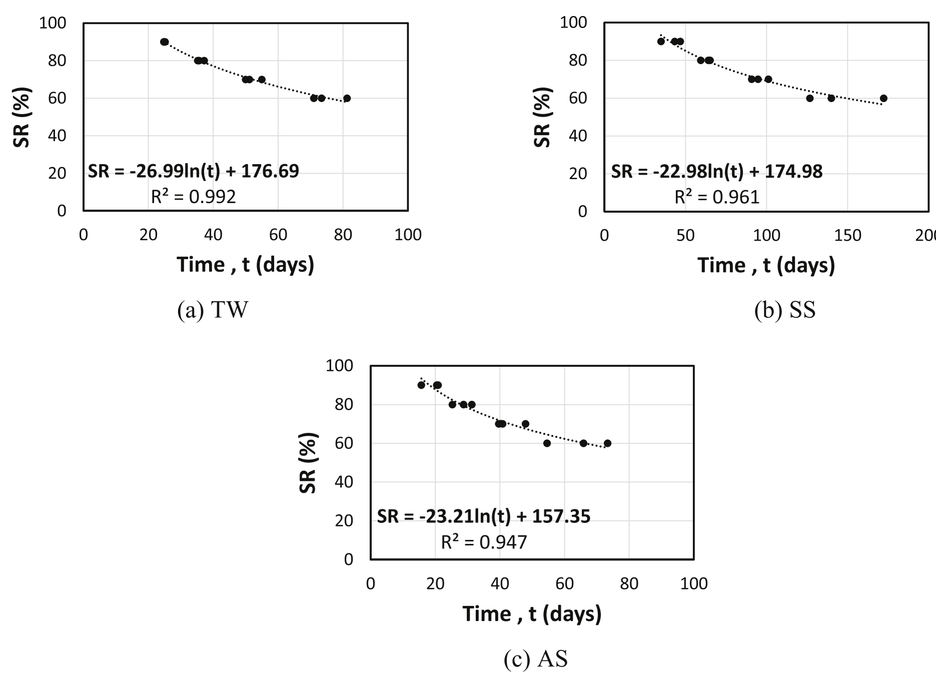


Fig. 12. Master curves for the bare bars at 23 °C.

bars in TW requires equating the relationships or the equations of the line derived in Fig. 12a for bare bars and in Fig. 13a for embedded bars. Simplifying these equations results in t_{ESL} for GFRP-reinforced concrete structures exposed in TW, as shown in Eq. (7), as follows:

$$t_{ESL-TW} = \frac{(TSF_{bb} \times t)^{7.606}}{TSF_{cb} \times 22 \times 10^{7}} \tag{7}$$

where TSF_{bb} and TSF_{cb} are the time-shift factors from the accelerated test and the concrete environment, respectively. Same procedure can be used to predict the equivalent service life for SS and AS as shown in Eqs. (7) and (8), respectively.

$$t_{ESL-SS} = \frac{(TSF_{bb} \times t)^{6.169}}{TSF_{cb} \times 49 \times 10^{6}} \tag{8}$$

$$t_{ESL-AS} = \frac{(TSF_{bb} \times t)^{5.080}}{TSF_{cb} \times 18 \times 10^{4}} \tag{9}$$

These equations correlate the exposure time needed in the laboratory for the bare GFRP bars to that of the equivalent service life of the GFRP bars embedded in concrete structures. For example, if an accelerated test

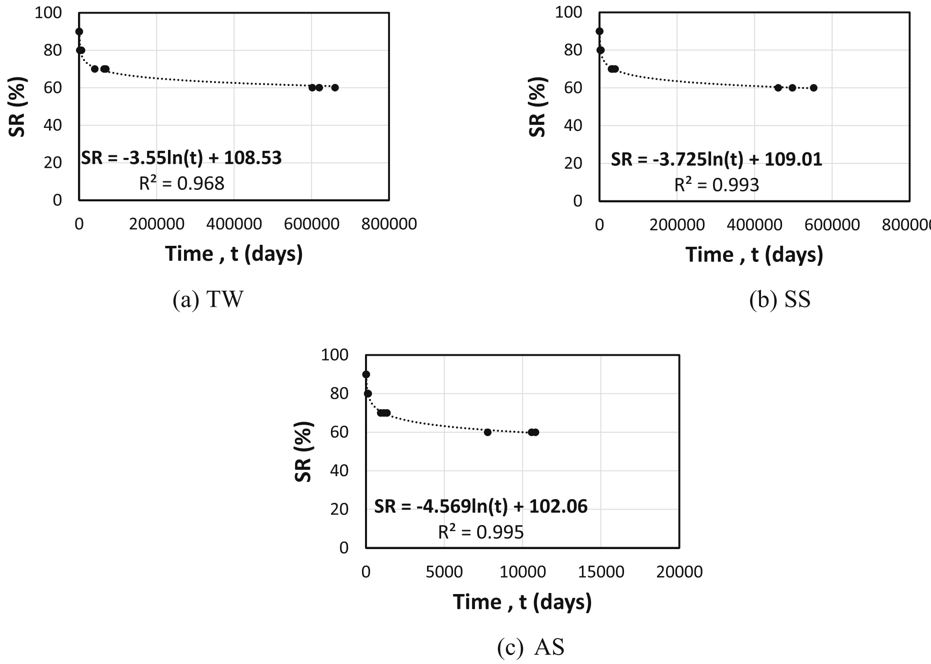


Fig. 13. Master curves for the embedded bars in concrete at 23 °C.

is conducted for GFRP bare bars immersed in alkaline solution at 60 °C for 56 days, the equivalent service life of the GFRP bar in actual concrete structure exposed to alkaline environment at 30 °C can be found by using Eq. (9) as follows:

$$t_{ESL-AS} = \frac{(2.33 \times 56)^{5.080}}{1.44 \times 18 \times 10^4} = 215441\ days = 590.2\ years$$

On the other hand, the predicted service life of GFRP bars in a concrete environment involved a mean annual temperature of 30 °C, which is the average annual temperature in Australia. This prediction of ILSS retention was performed as a function of a maximum service life of 100 years since bridge infrastructure in Australia is designed to be in service for this length of time [54]. Accordingly, master curves were created for the strength retention of GFRP bars in concrete structures during their service lives at a service temperature of 30 °C exposed to different environments (TW, SS, and AS) (see Fig. 14). These master curves were constructed according to the procedures described in developing Figs. 12 and 13, but using the corresponding time-shift factor (*TSF*) listed in Table 5. For example, the TSF values for bare and embedded GFRP bars exposed to alkaline solution at 30 °C are 1.10 and 1.44, respectively. Using these time-shift factors and the strength retention

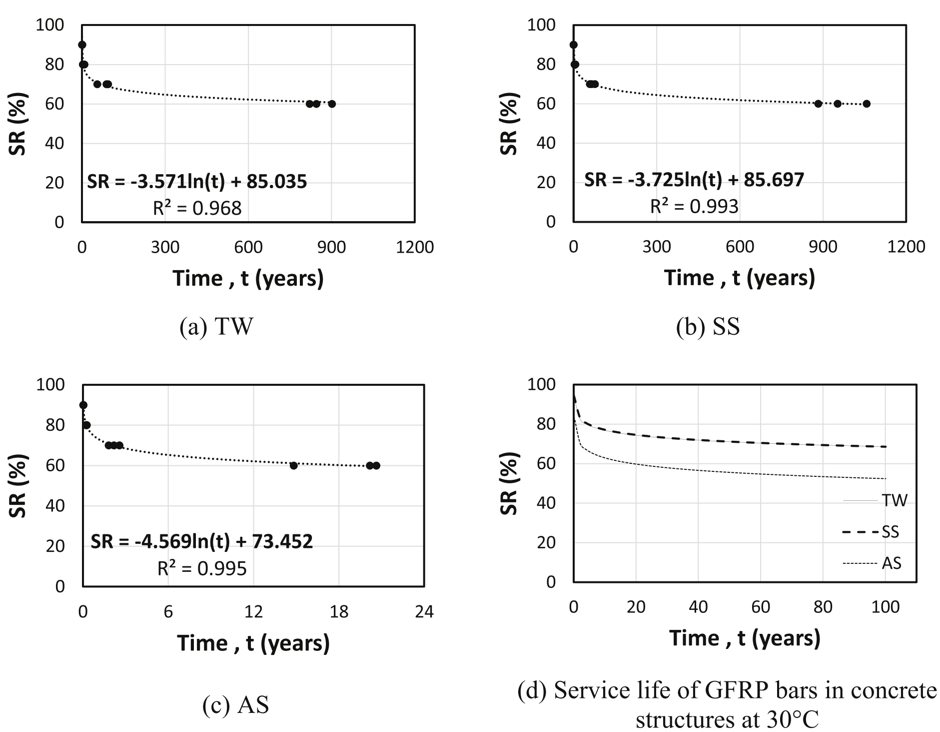


Fig. 14. Master curves for GFRP bars exposed to different environments at 30 °C.

equation in Eq. (6), the strength retention curve and the master curves for GFRP bars exposed to tap water, saline solution, and alkaline solution are derived and shown in Fig. 14. As can be seen from the graphs, the GFRP bars exposed to TW and SS could retain 60% of their ILSS for up to 900 years and 1050 years, respectively, while the GFRP bars in AS could retain 60% of their ILSS after 20 years in service. From the master curves in Fig. 14a-c, the SR of the GFRP bars in concrete structures exposed to different environments at an annual average temperature of 30 °C up to 100 years was established (see Fig. 14d).

In summary, the GFRP bars will retain 54% of their ILSS when exposed to alkaline environments (AS) and nearly 68% when exposed to tap water (TW) and seawater (SS) after 100 years of service (Fig. 14d). These retention values can also be conservatively applied for larger diameter bars, as Benmokrane et al. [1] found that the effect of conditioning on the ILSS of the GFRP bars was significant with the smaller diameter bars, but decreased as the diameter increased. This can be explained by the thinner layer affected by the alkaline solution in the case of the larger diameter bars. These results further show that the GFRP bars will last longer in a concrete environment than when directly exposed to simulated concrete conditions. These findings support the observations by Benmokrane et al. [24] who measured a maximum ILSS reduction of 16% for vinyl-ester GFRP bars extracted from concrete bridge barriers after 11 years in service. The slightly lower ILSS reduction in the field study could be due to the bars being well protected by a concrete cover of at least 65 mm. Similarly, the pH value measured from the core samples was only 12.3, which is lower than the pH of the cement used in our study. Nonetheless, these results clearly show that natural conditions are generally less aggressive to GFRP bars due to lower temperature and humidity conditions compared to accelerated exposure testing, which involves constant elevated temperature, continuous contact, and complete saturation in the solution. Finally, these findings confirm the conclusions of Wang et al. [29], who found that the long-term predictions for FRP bars placed directly in simulated solutions are too conservative when compared to field results for bars embedded in concrete.

6. Conclusions

This study comparatively evaluated the durability of GFRP bars in concrete and in a simulated concrete environment by investigating their interlaminar shear strength. It focused on evaluating the physical, mechanical, and microstructural properties of GFRP bars in high-moisture, saltwater, and alkali environments. The following conclusions can be drawn from this study:

- The percentage water uptake and the apparent diffusivity of the GFRP bars were strongly dependent on the type of solution and temperature, with the percentage water absorption and apparent diffusivity higher at higher temperatures. Given the same temperature exposure, the GFRP bars conditioned in the alkaline solution recorded the highest moisture uptake and apparent diffusivity rate, followed by the bars conditioned in tap water and then saline solution.
- The interlaminar shear strength of the GFRP bars decreased as the exposure temperature and duration increased. The ILSS of the concrete-embedded GFRP bars was generally higher than that of the bare GFRP bars subjected to similar immersion conditions.
- The alkaline solution was more aggressive to the GFRP bars than tap water and saline solution, affecting their interlaminar shear strength. After 112 days of conditioning at 60 °C, the bare GFRP bars exposed to this solution retained only 30% of their interlaminar shear strength, compared to 41% and 54%, respectively, for the bars exposed to tap water and saline solution.
- Direct immersion deteriorated the interlaminar shear strength of the GFRP bars more severely than immersion of bars embedded in concrete. After 112 days of conditioning at 80 °C, the concrete-embedded GFRP bars exposed to alkaline solution retained 68% of their interlaminar shear strength, compared to only 23% for the bare GFRP bars.
- SEM revealed that the fiber surface of the concrete-embedded GFRP bars had more resin coverage than the bare GFRP bars. Likewise, more residual resin was found on fiber surfaces in the case of the GFRP bars exposed to tap water or saline solution than those in alkaline solution, suggesting better bonding between the fibers and vinyl-ester resin.
- The FTIR spectra did not reveal any significant changes in the polymer's chemical structure, except for the relatively higher intensity of the OH stretching band for the GFRP bars directly immersed in the alkaline solution. This higher intensity was only observed at the bar surface, indicating that the water absorption was concentrated only in the thin resin-rich area of the GFRP bars.
- Based on the Arrhenius relation, the required activation energy to cause the degradation for GFRP bars in concrete was higher than with the bare GFRP bars immersed directly in different accelerated aging solutions, which explains the benefit of the surrounding concrete cover in extending the durability of the GFRP bars in actual structures.
- Master curves and time-shift factors to correlate the strength retention of the accelerated aging test of bare GFRP bars to the equivalent service life for GFRP bars in concrete environments were developed. Based on this correlation, the GFRP bars in actual concrete structures will retain up to 54%, 68%, and 68% of their ILSS after 100 years of service at an annual average temperature of 30 °C when exposed to an alkaline environment, tap water, or saline solution, respectively.

The results from this work provided a good representation and comparison of the long-term properties and durability performance of GFRP bars in simulated and actual concrete environments. Furthermore, the short-beam shear test gave a straightforward and reliable indication of the resistance of the fiber–matrix of GFRP bars exposed to different environmental conditions. A comparative study to relate the interfacial property of GFRP bars using the short-beam shear test to longitudinal properties could lead to a simpler, more practical assessment of the durability and long-term performance of GFRP bars in concrete environments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The first author acknowledges the scholarship granted by the Australian Government Endeavour Executive Leadership Award to undertake his professional development at the Centre for Integration of Composites into Infrastructure (CICI), the Natural Science and Engineering Research Council (NSERC) of Canada, West Virginia University, Morgantown, USA, and University of Sherbrooke, Sherbrooke, QC, Canada. The authors are grateful to Inconmat V-ROD Australia and Pultrall Inc. (Thetford Mines, QC, Canada) for providing the GFRP bars.

References

[1] B. Benmokrane, A. Manalo, J.-C. Bouhet, K. Mohamed, M. Robert, Effects of diameter on the durability of glass-fiber-reinforced-polymer (GFRP) bars conditioned in alkaline solution, ASCE J. Compos. Construct. 21 (5) (2017), 04017040-1:12.

[2] G.B. Maranan, A.C. Manalo, B. Benmokrane, W. Karunasena, P. Mendis, Behavior of concentrically loaded geopolymer-concrete circular columns reinforced longitudinally and transversely with GFRP bars, Eng. Struct. 117 (2016) 422–436.

[3] A.C. Manalo, B. Benmokrane, K. Park, D. Lutze, Recent developments on FRP bars as internal reinforcement in concrete structures, Concrete in Australia 40 (2) (2014) 46–56.
[4] A.C. Manalo, Behaviour of fibre composite sandwich structures under short and asymmetrical beam shear tests, Compos. Struct. 99 (2013) 339–349.
[5] A.C. Manalo, T. Aravinthan, W. Karunasena, In-plane shear behaviour of structural fibre composite sandwiches using asymmetrical beam shear test, Construct. Build. Mater. 24 (10) (2010) 1952–1960.
[6] W. Ferdous, A. Manalo, A. Khennane, O. Kayali, Geopolymer concrete-filled pultruded composite beams - concrete mix design and application, Cement Concr. Compos. 58 (2015) 1–13.
[7] J.W. Chin, K. Aouadi, M.R. Haight, W.L. Hughes, T. Nguyen, Effects of water, salt solution and simulated concrete pore solution on the properties of composite matrix resins used in civil engineering applications, Polym. Compos. 22 (2) (2001) 282–297.
[8] F.E. Tannous, H. Saadatmanesh, Durability of AR glass fiber reinforced plastic bars, ASCE J. Compos. Construct. 3 (1) (1999) 12–19.
[9] V.M. Karbhari, S. Zhang, "E-glass/vinyl ester composites in aqueous environments – I: experimental results", Appl. Compos. Mater. 10 (2003) 19–48.
[10] A. Belarbi, H. Wang, Durability of FRP bars embedded in fiber-reinforced concrete, ASCE J. Compos. Construct. 16 (4) (2012) 371–380.
[11] B. Benmokrane, A.H. Ali, H.M. Mohamed, A. ElSafty, A. Manalo, Laboratory assessment and durability performance of vinyl-ester, polyester, and epoxy glass-FRP bars for concrete structures, Composites Part B 114 (2017) 163–174.
[12] G. Nkurunziza, A. Debaiky, P. Cousin, B. Benmokrane, Durability of GFRP bars: a critical review of literature, Prog. Struct. Eng. Mater. 7 (2005) 194–209.
[13] J.F. Davalos, Y. Chen, I. Ray, Effect of FRP bar degradation on interface bond with high strength concrete, Cement Concr. Compos. 30 (8) (2008) 722–730.
[14] M. Robert, B. Benmokrane, Effect of aging on bond of GFRP bars embedded in concrete, Cement Concr. Compos. 32 (6) (2010) 461–467.
[15] F. Ceroni, E. Cosenza, M. Gaetano, M. Pecce, Durability issues of FRP bars in reinforced concrete members, Cement Concr. Compos. 28 (2006) 857–868.
[16] F. Micelli, A. Nanni, Durability of FRP rods for concrete structures, Construct. Build. Mater. 18 (2004) 491–503.
[17] O. Gooranorimi, A. Nanni, GFRP reinforcement in concrete after 15 Years of service, ASCE J. Compos. Construct. 21 (5) (2017), 04017024-1-9.
[18] V.M. Karbhari, G. Xian, Hygrothermal effects of high V_F pultruded unidirectional carbon/epoxy composites: moisture uptake, Composites Part B 40 (2009) 41–49.
[19] H. Kim, Y. Park, Y. You, Short-term durability test for GFRP rods under various environmental conditions, Compos. Struct. 83 (2008) 37–47.
[20] J.F. Davalos, Y. Chen, I. Ray, Long-term durability prediction models for GFRP bars in concrete environment, J. Compos. Mater. 46 (16) (2012) 1899–1914.
[21] T.H. Almusallam, Y.A. Al-Salloum, S.H. Alsayed, S. El-Gamal, M. Aqel, Tensile properties of glass fiber-reinforced polymer bars embedded in concrete under severe laboratory and field environmental conditions, J. Compos. Mater. 47 (4) (2012) 393–407.
[22] M. Robert, P. Cousin, B. Benmokrane, Durability of GFRP reinforcing bars embedded in moist concrete, ASCE J. Compos. Construct. 13 (2009) 66–73.
[23] A. Mufti, M. Onefrei, B. Benmokrane, N. Banhia, M. Boulfiza, J. Newhook, B. Bakht, G. Tadros, P. Brett, Field study of glass-fibre-reinforced polymer durability in concrete, Can. J. Civ. Eng. 34 (3) (2007) 355–366.
[24] B. Benmokrane, C. Nazair, M. Loranger, A. Manalo, Field durability of vinyl-ester-based GFRP rebars in concrete bridge barriers, ASCE J. Bridg. Eng. 23 (12) (2018), 04018094-1-13.
[25] T. D'Antino, M.A. Pisani, C. Poggi, Effect of the environment on the performance of GFRP reinforcing bars, Composites Part B 141 (2018) 123–136.
[26] M.A. Aiello, M. Leone, A.N. Aniskevich, O.A. Starkova, Moisture effects on elastic and viscoelastic properties of CFRP rebars and vinylester binder, ASCE J. Compos. Construct. 18 (2018) 686–691.
[27] H. Ashrafi, M. Bazli, A.V. Oskouei, L. Bazli, Effect of sequential exposure to UV radiation and water vapor condensation and extreme temperatures on the mechanical properties of GFRP bars, ASCE J. Compos. Construct. 22 (1) (2018), 04017047-1-17.
[28] D. Adams, What is the most important type of mechanical test for composites?, in: Composites World 2018, 2018, pp. 10–11, 4(1).
[29] Z. Wang, X.L. Zhao, G. Xian, G. Wu, R.K. Singh Raman, S. Al-Saadi, Durability study on interlaminar shear behaviour of basalt-, glass- and carbon-fibre reinforced polymer (B/G/CFRP) bars in seawater sea sand concrete environment, Construct. Build. Mater. 156 (2017) 985–1004.
[30] Canadian Standards Association (CSA), "Specification for Fibre-Reinforced polymers." *CAN/CSA-S807-10*, 2010. Rexdale, Ontario, Canada.
[31] International Standard ISO 1172, "Textile-glass-reinforced Plastics — Prepregs, Moulding Compounds and Laminates — Determination of the Textile-Glass and Mineral-Filler Content — Calcination Methods". ISO 1172:1996, ISO, Geneva, Switzerland, 1996.
[32] Canadian Standards Association (CSA), Design and Construction of Building Structures with Fibre-Reinforced polymers." *CAN/CSA-S806:2012:R2017*, 2017. Rexdale, Ontario, Canada.
[33] ASTM Standard ASTM D3171, Standard Test Methods for Constituent Content of Composite Materials. ASTM D3171-15, ASTM International, West Conshohocken, Philadelphia, Pa, 2015, 19103.
[34] ASTM Standard ASTM E1131, Standard Test Method for Compositional Analysis by Thermogravimetry. ASTM E1131 – 08 (2014), ASTM International, West Conshohocken, Philadelphia, Pa, 2014, 19103.
[35] ASTM Standard ASTM D5117, Standard Test Method for Dye Penetration of Solid Fiberglass Reinforced Pultruded Stock. ASTM D5117 – 17, ASTM International, West Conshohocken, Philadelphia, Pa, 2017, 19103.
[36] ASTM Standard ASTM D570, Water Absorption of Plastics. ASTM D570-18, ASTM International, West Conshohocken, Philadelphia, Pa, 2018, 19103.
[37] ASTM Standard ASTM E1356, Standard Test Method for Assignment of the Glass Transition Temperatures by Differential Scanning Calorimetry. ASTM E1356-08 (2014), ASTM International, West Conshohocken, Philadelphia, Pa, 2014, 19103.
[38] ASTM Standard ASTM D4475 -02, Standard Test Method for Apparent Horizontal Shear Strength of Pultruded Reinforced Plastic Rods by the Short-Beam Method. ASTM D4475-02 (2016), ASTM International, West Conshohocken, Philadelphia, Pa, 2016, 19103.
[39] ASTM Standard ASTM C150/C150M, Standard Specification for Portland Cement. *ASTM C150/C150M – 19a*, ASTM International, West Conshohocken, Philadelphia, Pa, 2019, 19103.
[40] ASTM Standard ASTM D4972, Standard Test Method for pH of Soils. ASTM D4972-13, ASTM International, West Conshohocken, Philadelphia, Pa, 2013, 19103.
[41] L.C. Bank, R.T. Gentry, B.P. Thompson, S.J. Russell, A model specification for FRP composites for civil engineering structures, Construct. Build. Mater. 17 (6–7) (2003) 405–437.
[42] B. Benmokrane, E. El-Salakawy, S. El-Gamal, S. Goulet, Construction and testing of an innovative concrete bridge deck totally reinforced with glass FRP bars: Val-Alain bridge on highway 20 east, J. Bridge Eng. 12 (5) (2007) 632–645.
[43] ASTM Standard ASTM D 1141-98, Standard Practice for the Preparation of Substitute Ocean Water. *ASTM D1141-98*, ASTM International, West Conshohocken, Philadelphia, Pa, 2013, 19103.
[44] ASTM Standard ASTM D7705/D7705M - 12: Standard Test Method for Alkali Resistance of Fiber Reinforced Polymer (FRP) Matrix Composite Bars Used in Concrete Construction. ASTM D7705/D7705M-12, ASTM International, West Conshohocken, Philadelphia, Pa 19103.
[45] V.M. Karbhari, G. Xian, Hygrothermal effects of high V_F pultruded unidirectional carbon/epoxy composites: moisture uptake, Composites Part B 40 (2009) 41–49.
[46] Y. Chen, J.F. Davalos, I. Ray, Durability prediction for GFRP reinforcing bars using short-term data of accelerated aging tests, J. Compos. Construct. 10 (4) (2006) 279–286.
[47] D.E. Mouzakis, H. Zoga, C. Galiotis, Accelerated environmental ageing of polyester/glass fiber reinforced composites (GFRPCs), Composites Part B 39 (2008) 467–475.
[48] M. Robert, B. Benmokrane, Combined effects of saline solution and moist concrete on long-term durability of GFRP reinforcing bars, Construct. Build. Mater. 38 (2013) 274–284.
[49] S. Naya, A. Meneses, J. Tarrío-Saavedra, R. Artiaga, J. López-Beceiro, C. Gracia-Fernández, New method for estimating shift factors in time–temperature superposition models, J. Therm. Anal. Calorim. 113 (2) (2013) 453–460.
[50] W.B. Nelson, Accelerated Testing: Statistical Models, Test Plans, and Data Analysis, John Wiley & Sons, 2009.
[51] V. Dejke, Durability of FRP Reinforcement in Concrete: Literature Review and Experiments, Chalmers University, Sweden, 2001.
[52] A.H. Ali, H.M. Mohamed, B. Benmokrane, A. ElSafty, O. Chaallal, Durability performance and long-term prediction models of sand-coated basalt FRP bars, Compos. B Eng. 157 (2019) 248–258.
[53] A.H. Ali, H.M. Mohamed, B. Benmokrane, A. ElSafty, O. Chaallal, Theory-based approaches and microstructural analysis to evaluate the service life retention pf stressed carbon fiber composite strands for concrete bridge applications, Compos. B Eng. 165 (2019) 279–292.
[54] Austroads, Realising 100-years Bridge Design Life in an Aggressive Environment: Review of the Literature. Austroads Technical Report AP-T313-16, 2016.